

BENTLEY
AutoPIPE Advanced 12.06.00.48

AKABEL._ASME B31.8
08/07/2022
09:46 PM

BENTLEY
AutoPIPE Advanced 12.06.00.48

**
** AUTOPIPE SYSTEM INFORMATION **
**

SYSTEM NAME : AKABEL._ASME B31.8

PROJECT ID : AKABEL

PREPARED BY : _____
I.O

CHECKED BY : _____
E N

1ST APPROVER : _____

2ND APPROVER : _____

PIPING CODE : ASME B31.8

YEAR : 2016

VERTICAL AXIS : Y

AMBIENT TEMPERATURE : 24.0 deg C

COMPONENT LIBRARY : AUTOPIPE

MATERIAL LIBRARY : B318-16

MODEL REVISION NUMBER : 14

*** Model changed and analysis results are outdated. Please re-analyze ***

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C O O R D I N A T E S D A T A L I S T I N G

POINT NAME	-----COORDINATE (mm)----- X Y Z	Mass Points	Piping Restraint
*** SEGMENT A, LINE # AKABP-36"-PL-012-X70-CL900 UG			
A00	-562000.00 4940.00 -0.02	1115	Unrestrained
A01	-4000.00 4940.00 0.00	3	Unrestrained
A02 N	-2214.11 4940.00 0.00	1	Unrestrained
A02	-2000.00 4940.00 0.00	0	Unrestrained
A02 F	-1796.08 5005.26 0.00	3	Unrestrained
A03	0.00 5580.00 0.00	4	Unrestrained
A04	2000.00 6220.00 0.00	0	Unrestrained
A05 N	5796.08 7434.75 0.00	0	Unrestrained
A05	6000.00 7500.00 0.00	0	Unrestrained
A05 F	6214.11 7500.00 0.00	0	Unrestrained
A06	7500.00 7500.00 0.00	0	Unrestrained
A07	9000.00 7500.00 0.00	0	Unrestrained
A08	11600.00 7500.00 0.00	0	Unrestrained
A08 M	12743.00 7500.00 0.00	0	Unrestrained
A09	13886.00 7500.00 0.00	0	Unrestrained
A10	17182.80 7500.00 0.00	0	Unrestrained
A11	18382.80 7500.00 0.00	0	Unrestrained
A11 M	19525.80 7500.00 0.00	0	Unrestrained
A12	20668.80 7500.00 0.00	0	Unrestrained
A13	23965.60 7500.00 0.00	0	Unrestrained
A13 M	25108.60 7500.00 0.00	0	Unrestrained
A14	26251.60 7500.00 0.00	0	Unrestrained
A15	27248.40 7500.00 0.00	0	Unrestrained
A16	29848.40 7500.00 0.00	0	Unrestrained
A17	31098.40 7500.00 0.00	0	Unrestrained
A18	32348.40 7500.00 0.00	0	Unrestrained
A19	34798.40 7500.00 0.00	0	Unrestrained
A19 M	35941.40 7500.00 0.00	0	Unrestrained
A20	37084.40 7500.00 0.00	0	Unrestrained
A21	36931.20 7500.00 0.00	0	Unrestrained
*** SEGMENT B, LINE # AKAB-A-100			
A21	36931.20 7500.00 0.00	0	Unrestrained
B01	38231.20 7500.00 0.00	0	Unrestrained
B02	39631.20 7500.00 0.00	0	Unrestrained
B03	43631.20 7500.00 0.00	0	Unrestrained
B04	44251.20 7500.00 0.00	0	Unrestrained
B05	44851.20 7500.00 0.00	0	Unrestrained
B06	47751.20 7500.00 0.00	0	Unrestrained
B07	49451.20 7500.00 0.00	0	Unrestrained
B08	50651.20 7500.00 0.00	0	Unrestrained
*** SEGMENT C, LINE # AKAB-A-100_N9			
B05	44851.20 7500.00 0.00	0	Unrestrained
C01	44851.20 8200.00 0.00	0	Unrestrained
*** SEGMENT F, LINE # AKABP-36"-PL-012-X70-CL900_B			
A15	27248.40 7500.00 0.00	0	Unrestrained
F01 N	27248.40 8928.40 0.00	0	Unrestrained
F01	27248.40 10300.00 0.00	0	Unrestrained
F01 F	27248.40 10300.00 -1371.60	0	Unrestrained

C O O R D I N A T E S D A T A L I S T I N G

POINT NAME	-----COORDINATE (mm) ----- X Y Z			Mass Points	Piping Restraint
F02 N	27248.40	10300.00	-2428.40	0	Unrestrained
F02	27248.40	10300.00	-3800.00	0	Unrestrained
F02 F	27248.40	8928.40	-3800.00	0	Unrestrained
F03 N	27248.40	8871.60	-3800.00	0	Unrestrained
F03	27248.40	7500.00	-3800.00	0	Unrestrained
F03 F	25876.80	7500.00	-3800.00	0	Unrestrained
F04	25648.40	7500.00	-3800.00	0	Unrestrained
F04 M	24505.40	7500.00	-3800.00	0	Unrestrained
F05	23362.40	7500.00	-3800.00	0	Unrestrained
F06	19362.40	7500.00	-3800.00	0	Unrestrained
F07 N	18554.40	7500.00	-3800.00	0	Unrestrained
F07	17182.80	7500.00	-3800.00	0	Unrestrained
F07 F	17182.80	8871.60	-3800.00	0	Unrestrained
F08 N	17182.80	8928.40	-3800.00	0	Unrestrained
F08	17182.80	10300.00	-3800.00	0	Unrestrained
F08 F	17182.80	10300.00	-2428.40	0	Unrestrained
F09 N	17182.80	10300.00	-1371.60	0	Unrestrained
F09	17182.80	10300.00	0.00	0	Unrestrained
F09 F	17182.80	8928.40	0.00	0	Unrestrained
A10	17182.80	7500.00	0.00	0	Unrestrained
*** SEGMENT G, LINE # AKAB-A-100_N4					
B02	39631.20	7500.00	0.00	0	Unrestrained
G01	39631.20	8200.00	0.00	0	Unrestrained
*** SEGMENT I, LINE # AKAB-A-100_N7					
B06	47751.20	7500.00	0.00	0	Unrestrained
I01	47751.20	8200.00	0.00	0	Unrestrained

Pipe ID/ Material	Nom/ O.D. Sch mm	---Thickness(mm)--- W.Th. Corr Mill	Insu Ling	Spec Grav/ InsMt	InsuDen/ LingDen/ CladDen kg/m3	Weight(N/m) Pipe/ Ling/ Total Cont Insu/ Clad	ZL/ ZC	Composition/ ZC
CladMaterial		Clad						
---Line Class---								

Tag No. :	<None>												
4"B06	100	114.30	11.13	6.00	1.39	0	0	0	0.000	277	0	277	1.00
5LS-B	120								0.000	0	0		1.00
			0						0.000		0		
150													

Pipe ID/ Material	Nom/ O.D. Sch mm	-----Thickness(mm)----- W.Th. Corr Mill Insu Ling	Spec Grav/	InsuDen/ LingDen/	Weight(N/m) Pipe/ Ling/ Total	ZL/ ZC	Composition/
CladMaterial		Clad	InsMt	CladDen	Cont Insu/		
---Line Class---				kg/m3	Clad		

Tag No. :	<None>												
2"B06	50	60.325	11.07	6.00	1.38	0	0	0	0.000	132	0	132	1.00
5LS-B	XXS								0.000	0	0		1.00
			0						0.000		0		
150													

M A T E R I A L D A T A L I S T I N G

Material Name	Pipe ID	Density kg/m3	Pois. Ratio	Temper. deg C	Modulus E6 Axial	N/mm2 Hoop	Expans. mm/m	Composition
5LX-X70	36"CL900	7833.0	0.30	24.0	0.20320	0.20320	0.07815	
				60.0	0.20080			0.4154
				12.7	0.20382			-0.1280
				23.0	0.20327			-0.0115
				100.0	0.19816			0.8900
				-29.0	0.20593			-0.5971
5LX-X70	40"PIPE	7833.0	0.30	24.0	0.20320	0.20320	0.07815	
				100.0	0.19816			0.8900
				-29.0	0.20593			-0.5971
				23.0	0.20327			-0.0115
5L-B	12"C06	7833.0	0.30	24.0	0.20320	0.20320	0.07815	
				100.0	0.19816			0.8900
				-29.0	0.20593			-0.5971
				23.0	0.20327			-0.0115
5LS-B	4"C06	7833.0	0.30	24.0	0.20320	0.20320	0.07815	
				100.0	0.19816			0.8900
				-29.0	0.20593			-0.5971
				23.0	0.20327			-0.0115

M A T E R I A L A L L O W A B L E D A T A L I S T I N G

Material Name	Pipe ID	Temper. deg C	Yield N/mm2
5LX-X70	36"CL900	24.0	482.63
5LX-X70	40"PIPE	24.0	482.63
5L-B	12"C06	24.0	241.32
5LS-B	4"C06	24.0	241.32

OPERATING TEMPERATURE AND PRESSURE DATA
STRESSES IN N/mm2

POINT NAME	CASE	PRESS. N/mm2	TEMPER deg C	EXPAN. mm/m	MODULUS E6 N/mm2	YIELD STRESS
*** SEGMENT A, LINE # AKABP-36"-PL-012-X70-CL900_UG						
A00	T1	11.0000	60.00	0.415	0.20080	482.63
	T2	11.0000	12.70	-0.128	0.20382	482.63
	T3	11.0000	23.00	-0.012	0.20327	482.63
A21	Same as previous point.					
*** SEGMENT B, LINE # AKAB-A-100						
A21	T1	11.0000	100	0.890	0.19816	482.63
	T2	11.0000	-29.00	-0.597	0.20593	482.63
	T3	11.0000	23.00	-0.012	0.20327	482.63
B08	Same as previous point.					
*** SEGMENT C, LINE # AKAB-A-100 N9						
B05	T1	11.0000	100	0.890	0.19816	241.32
	T2	11.0000	-29.00	-0.597	0.20593	241.32
	T3	11.0000	23.00	-0.012	0.20327	241.32
C01	Same as previous point.					
*** SEGMENT F, LINE # AKABP-36"-PL-012-X70-CL900_B						
A15	T1	11.0000	60.00	0.415	0.20080	482.63
	T2	11.0000	12.70	-0.128	0.20382	482.63
	T3	11.0000	23.00	-0.012	0.20327	482.63
A10	Same as previous point.					
*** SEGMENT G, LINE # AKAB-A-100 N4						
B02	T1	11.0000	100	0.890	0.19816	241.32
	T2	11.0000	-29.00	-0.597	0.20593	241.32
	T3	11.0000	23.00	-0.012	0.20327	241.32
G01	Same as previous point.					
*** SEGMENT I, LINE # AKAB-A-100 N7						
B06	T1	11.0000	100	0.890	0.19816	241.32
	T2	11.0000	-29.00	-0.597	0.20593	241.32
	T3	11.0000	23.00	-0.012	0.20327	241.32
I01	Same as previous point.					

u User-defined value

* Non-code material for allowable stress;
Non-standard material for expansion and modulus

SEGMENT DATA LISTING

Segment Name	First Node	Last Node	Line Number	Apply Wind	Apply Bowling	Apply Buoyancy
A	A00	A21	AKABP-36"-PL-012-X70-CL900_UG	No	No	No
B	A21	B08	AKAB-A-100	Yes	No	No
C	B05	C01	AKAB-A-100_N9	Yes	No	No
F	A15	A10	AKABP-36"-PL-012-X70-CL900_B	Yes	No	No
G	B02	G01	AKAB-A-100_N4	Yes	No	No
I	B06	I01	AKAB-A-100_N7	Yes	No	No

SOIL DATA LISTING

SOIL STIFFNESS PROPERTIES (SAND36)

Dirn	Auto	Initial K (N/mm/mm)	Auto	Yield P (N/m)	Final K (N/mm/mm)	Yield disp (mm)
-----	-----	-----	-----	-----	-----	-----
Low Stiffness						
Horiz.	Y	3.875	Y	307121.19	0.000	79.2480
Long.	Y	11.481	Y	29162.150	0.000	2.5400
Vert. Up	Y	2.176	Y	33167.953	0.000	15.2400
Vert. Dn	Y	23.265	Y	2127338.8	0.000	91.4400
High Stiffness						
Horiz.	Y	3.875	Y	307121.19	0.000	79.2480
Long.	Y	11.481	Y	29162.150	0.000	2.5400
Vert. Up	Y	2.176	Y	33167.953	0.000	15.2400
Vert. Dn	Y	23.265	Y	2127338.8	0.000	91.4400
Average Stiffness						
Horiz.	Y	3.875	Y	307121.19	0.000	79.2480
Long.	Y	11.481	Y	29162.150	0.000	2.5400
Vert. Up	Y	2.176	Y	33167.953	0.000	15.2400
Vert. Dn	Y	23.265	Y	2127338.8	0.000	91.4400

SOIL PARAMETERS (SAND36)

Calculation Method : American Lifeline Alliance (ASCE 2001)
Soil Type : Dense Sand
Pipe Direction : Horizontal

Parameters	Low	High	Average
Outside Diameter, D [mm]	914.40		
Depth to Centerline, H [mm]	1524.00		
Effective Unit Wt. above pipe [kg/m3]	1601.85	1601.85	1601.85
Total Unit Wt. below pipe [kg/m3]	1601.85	1601.85	1601.85
Dry Unit Wt. above pipe [kg/m3]	1601.85	1601.85	1601.85
Soil Cohesion, c [N/m2]	0.00	0.00	0.00
Friction Angle, phi [deg]	40.00	40.00	40.00
Coating Factor, f	0.80	0.80	0.80
Coefficient of pressure at rest, K0	0.36	0.36	0.36
Horizontal Yield Displacement, dp [mm]	79.25	79.25	79.25
Longitudinal Yield Displacement, dt [mm]	2.54	2.54	2.54
Vertical Up Yield Displacement, dqu [mm]	15.24	15.24	15.24
Vertical Dn Yield Displacement, dqd [mm]	91.44	91.44	91.44
*Height of water on top of pipe [mm]	0.00	0.00	0.00
*Soil Adhesion, Ca [N/m2]	47880.00	47880.00	47880.00

Note: Fields with * are used in soil overburden calculations only and do not effect soil stiffness calculations.

Computed soil parameters (ASCE Method):

Longitudinal	Adhesion alpha	0.00	0.00	0.00
	Pipe/Soil delta=f*phi [deg]	32.00	32.00	32.00
Horizontal	Nch	0.00	0.00	0.00
	Nqh	14.03	14.03	14.03
Vertical Up	Ncv	0.00	0.00	0.00
	Nqv	1.52	1.52	1.52
	Soil Weight on top Ws [N/m]	21890.85	21890.85	21890.85
Vertical Down	Nc	75.32	75.32	75.32
	Nq	64.20	64.20	64.20
	Ngamma	109.95	109.95	109.95

Derived values for buried piping :

Circumferential wall bending stress method:	Adams et. al.			
Circumferential wall bending stress [N/mm2]	17.12	17.12	17.12	
Linear buoyancy force on pipe, Fb [N/m]	0.00	0.00	0.00	
Equiv. seismic wave prop. temperature [deg C]	-17.78	-17.78	-17.78	
Buoyancy reduction factor R	1.00	1.00	1.00	

Burial factor B' (Adams method - for Pipe ID 36"CL900)	0.21
Burial factor B' (CC N-755-1 - for Pipe ID 36"CL900)	0.24

VIRTUAL ANCHOR DATA (SAND36)

Pipe Identifier : 36"CL900
Point name for temp. data : A00
Operating Case : OP1
Thermal Expansion : 0.415 mm/m
Temperature Change : 36.000 deg C
Operating Pressure : 11.000 N/mm2
Virtual Anchor lengths Lm : 279713.13 mm
La : 559426.25 mm
Lb : 14337.039 mm

EARTH LOADS (SAND36)

Pipe Identifier : 36"CL900
Surface live load file name :
Pressure on pipe due to surface load Pp : 0.000 N/mm2
Trench laying condition : Type 1
Modulus of passive soil fill reaction E' : 1.034 N/mm2
Bending moment coefficient Kb : 0.235
Deflection coefficient Kx : 0.108
Deflection lag factor D1 : 1.500
Bedding constant K : 0.100
Soil support factor Fs : 0.100

SEISMIC LOADS (SAND36)

Pipe Identifier : 36"CL900
Soil input parameters set : Low
Seismic wave type : Compression
Seismic wave coefficient aw : 1.000
Seismic wave curvature coefficient ak : 1.600
Wave length of seismic wave Lw : 2540.000 mm
Seismic wave velocity at pipeline c : 1066800.0 mm/s
Peak ground acceleration PGA : 2540.000 mm/s2
Peak ground velocity PGV : 381.000 mm/s
Add axial and bending strain : N

UPHEAVAL BUCKLING (SAND36)

Consider Upheaval Buckling : N
Use K. Peters Compressive Force Calculation : N

L O A D S S U M M A R Y D A T A L I S T I N G

WIND LOAD CASES :

Number of load cases : 4

Ground elevation for wind : 2000.00 mm

Wind shape factor multiplier for piping: 1.000

Wind shape factor multiplier for beams: 1.000

Exposure factor for buried pipes : 0.000

Exposure factor for submerged pipes : 0.000

Exposed Segments - B C F G I

Wind application method : Projected

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 1 - W1

Direction - Global X-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"

Default Gust Factor G assumes pipe frequency > 1 Hz.

Piping is assumed not to be on an isolated hill (Kzt=1).

Automatic Cf reported in chart above is calculated

assuming D=1 ft. Program uses actual Pipe OD to

calculate Cf at height during analysis.

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 1 - W1

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 2 - W2

Direction - Global -X-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"

Default Gust Factor G assumes pipe frequency > 1 Hz.

Piping is assumed not to be on an isolated hill (Kzt=1).

Automatic Cf reported in chart above is calculated

assuming D=1 ft. Program uses actual Pipe OD to

calculate Cf at height during analysis.

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 2 - W2

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 3 - W3

Direction - Global Z-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"

Default Gust Factor G assumes pipe frequency > 1 Hz.

Piping is assumed not to be on an isolated hill (Kzt=1).

Automatic Cf reported in chart above is calculated

assuming D=1 ft. Program uses actual Pipe OD to

calculate Cf at height during analysis.

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 3 - W3

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 4 - W4

Direction - Global -Z-axis

ASCE 7-2010

Basic wind speed at 33 ft (10m): 136.79 kmph

Exposure category : B

Gust effect factor (G) : 0.85

Force coefficient (Cf), Figure 29.5-1 : Automatic

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (PIPING)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	0.700	508.965	302.834
3048.0	0.575	0.700	508.965	302.834
6096.0	0.624	0.700	552.567	328.777
10058.4	0.720	0.700	637.563	379.350
15240.0	0.811	0.700	717.930	427.168
22860.0	0.910	0.700	806.109	479.635
30480.0	0.988	0.700	875.166	520.724
60960.0	1.205	0.700	1066.839	634.770
152400.0	1.565	0.700	1386.105	824.733
304800.0	1.908	0.700	1689.681	1005.360

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

Per ASCE Figure 29.5-1, wind appl. method should be "Projected"

Default Gust Factor G assumes pipe frequency > 1 Hz.

Piping is assumed not to be on an isolated hill (Kzt=1).

Automatic Cf reported in chart above is calculated

assuming D=1 ft. Program uses actual Pipe OD to

calculate Cf at height during analysis.

L O A D S S U M M A R Y D A T A L I S T I N G

Load case 4 - W4

Structural members wind load application data

Structural members wind facing width: Wind facing

Gust effect factor (G) for structures (manual): 0.85

Force coefficient (Cf) for structures (manual): 2.000

ASCE 7-10 PARAMETERS AS A FUNCTION OF HEIGHT (STRUCTURE)

HEIGHT (mm)	Kz	Cf	Qz (N/m2)	Pressure (N/m2)
0.6	0.575	2.000	508.965	865.241
3048.0	0.575	2.000	508.965	865.241
6096.0	0.624	2.000	552.567	939.364
10058.4	0.720	2.000	637.563	1083.857
15240.0	0.811	2.000	717.930	1220.481
22860.0	0.910	2.000	806.109	1370.385
30480.0	0.988	2.000	875.166	1487.782
60960.0	1.205	2.000	1066.839	1813.627
152400.0	1.565	2.000	1386.105	2356.379
304800.0	1.908	2.000	1689.681	2872.458

Kz = Velocity pressure exposure coefficient

Cf = Force coefficient or wind shape factor

Qz = Velocity pressure = $0.00256 \cdot Kz \cdot Kzt \cdot V^2$ (Eq. 27.3-1)

Pressure = $Qz \cdot G \cdot Cf$ (Eq. 23)

S U P P O R T D A T A L I S T I N G

Point Name	Support Type	Support ID	Conn.to /Dir	Comp.Wt (kg)	Stiff.	Gap 1 (mm)	Gap 2 (mm)	Fric. Fact.	GapSet /#hgr	Preload (N)	Ld.Var	Size	Figure
Tag No.: <None>													
A08	M Shoe	A08 M1	Ground	0.000	Rigid	0.0	2540.0	0.00	Weightless				
Attachment ID: <None>													
Tag No.: <None>													
A11	M Shoe	A11 M1	Ground	0.000	Rigid	0.0	2540.0	0.00	Weightless				
Attachment ID: <None>													
Tag No.: <None>													
A13	M Shoe	A13 M1	Ground	0.000	Rigid	0.0	2540.0	0.00	Weightless				
Attachment ID: <None>													
A17	Guide-V	A17 1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless				
	Guide-H	A17 1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless				
Attachment ID: <None>													
Tag No.: <None>													
A19	M Shoe	A19 M1	Ground	0.000	Rigid	0.0	2540.0	0.00	Weightless				
Attachment ID: <None>													
Tag No.: <None>													
B01	Shoe	B01 1	Ground	0.000	Rigid	0.0	2540.0	0.10	Weightless				
Attachment ID: <None>													
B07	Guide-V	B07 1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless				
	Guide-H	B07 1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless				
Attachment ID: <None>													
Tag No.: <None>													
F04	M Shoe	F04 M1	Ground	0.000	Rigid	0.0	2540.0	0.00	Weightless				
Attachment ID: <None>													
Tag No.: <None>													
F06	Guide-V	F06 1	Ground	0.000	Rigid	0.0	3.0	0.10	Weightless				
	Guide-H	F06 1	Ground	0.000	Rigid	3.0	3.0	0.10	Weightless				

Spring Manufacturer: Anvil/Grinnell

NOTE 1: Soil supports are present but not listed.

Gap 1 : V-stop,Guide-V=down, Linestop,Incline,Tie/link=backward, Guide-H=Left

Gap 2 : V-stop,Guide-V=Up , Linestop,Incline,Tie/link=forward , Guide-H=Right

Stiffness units for rotation support: N.m/deg , all others: N/mm

B E N D D A T A L I S T I N G

Point Name	Bend Type	Radius (mm)	Angle (deg)	OD (mm)	tnom (mm)	Material	Flan^ /Cuts	SIF in	SIF out	Analysis Set	Flexibility (K)	Pressure Case
A02	Elbow	1372L	17.7	914.4	23.8	5LX-X70	0/0	2.99	2.49	1	10.01	None
								2.99	2.49	2	10.01	None
A05	Elbow	1372L	17.7	914.4	23.8	5LX-X70	0/0	2.99	2.49	1	10.01	None
								2.99	2.49	2	10.01	None
F01	Elbow	1372L	90.0	914.4	23.8	5LX-X70	0/0	2.99	2.49	1	10.01	None
								2.99	2.49	2	10.01	None
F02	Elbow	1372L	90.0	914.4	23.8	5LX-X70	0/0	2.99	2.49	1	10.01	None
								2.99	2.49	2	10.01	None
F03	Elbow	1372L	90.0	914.4	23.8	5LX-X70	1/0	2.22	1.85	1	7.41	None
								2.22	1.85	2	7.41	None
F07	Elbow	1372L	90.0	914.4	23.8	5LX-X70	0/0	2.99	2.49	1	10.01	None
								2.99	2.49	2	10.01	None
F08	Elbow	1372L	90.0	914.4	23.8	5LX-X70	0/0	2.99	2.49	1	10.01	None
								2.99	2.49	2	10.01	None
F09	Elbow	1372L	90.0	914.4	23.8	5LX-X70	0/0	2.99	2.49	1	10.01	None
								2.99	2.49	2	10.01	None

^ = Number of bend ends where either a flange or valve is within distance
(L/D) x Nominal diameter, where L/D is defined under Tools > Model options > Edit
L = Long radius

T E E D A T A L I S T I N G

Point Name	Point Seg-Type	Pipe OD (mm)	Pipe Thk. (mm)	Material	Tee Type	**Crotch Radius (mm)	*Crotch Thick (mm)	SIF in	SIF out
A10	A-Header F-Branch	914.4 914.4	23.8 23.8	5LX-X70 5LX-X70	Welding			2.02	2.36
A15	A-Header F-Branch	914.4 914.4	23.8 23.8	5LX-X70 5LX-X70	Welding			2.02	2.36
B02	B-Header G-Branch	914.4 114.3	23.8 17.1	5LX-X70 5LS-B	Other			1.00	1.00
B05	B-Header C-Branch	1016.0 323.9	24.0 28.6	5LX-X70 5L-B	Other			1.00	1.00
B06	B-Header I-Branch	1016.0 114.3	24.0 17.1	5LX-X70 5LS-B	Other			1.00	1.00

* = Pad thickness for reinforced tee.

**= Fillet radius for Raised tee or discontinuity dist. for Thickened tee.

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V A L V E D A T A L I S T I N G

Point		Dnom	Length	Valve	Auto	Symbol	Actuator	DX	DY	DZ		Joint	Size	Type
Name	Type	(mm)	(mm)	Weight			Weight	(mm)	(mm)	(mm)	SAF	Type	/Avg.	/Max.
				(kg)			(kg)						(mm)	(mm)
Tag No. : SDV1003														
A08	CATALOG	900	2286.00	4838	Yes	No					0.00	WN	1.00	
Tag No. : MV1043														
A11	CATALOG	900	2286.00	4838	Yes	No					0.00	WN	1.00	
Tag No. : MV1045														
A13	CATALOG	900	2286.00	4838	Yes	No					0.00	WN	1.00	
Tag No. : MV1057														
A19	CATALOG	900	2286.00	4838	Yes	No					0.00	WN	1.00	
Tag No. : MV1044														
F04	CATALOG	900	2286.00	4838	Yes	No					0.00	WN	1.00	

C A T A L O G V A L V E D A T A L I S T I N G

Point Name : A08
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : A11
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : A13
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : A19
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

Point Name : F04
Manufacturer: AutoPIPE Generic
Standard : ANSI/ASME
Sub Category: Ball Valves
Type : Ball Valve Side Entry
Main End : FL
Rating : 900

F L A N G E D A T A L I S T I N G

Point	FLG		Dnom	Rating	Flange	Bolt	Joint		Size	Type	ANSI
Name	No.	Type	(mm)		Weight	Weight	Type	SIF	/Avg.	/Max.	Check
			(mm)		(kg)	(kg)			(mm)	(mm)	
Tag No. :		<None>									
A08	1	WELDNECK	900		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
A09	1	WELDNECK	900		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
A11	1	WELDNECK	900		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
A12	1	WELDNECK	900		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
A13	1	WELDNECK	900		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
A14	1	WELDNECK	900		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
A19	1	WELDNECK	900		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
A20	1	WELDNECK	900		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
B08	1	BLIND	1000		9001400.00	700.00	USER	1.00			No
Tag No. :		<None>									
C01	1	WELDNECK	300		900 168.74	5.45	WN	1.00			No
Tag No. :		<None>									
C01	2	WELDNECK	300		900 168.74	5.45	WN	1.00			No
Tag No. :		<None>									
F04	1	WELDNECK	900		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
F05	1	WELDNECK	900		9001764.00	882.00	WN	1.00			No
Tag No. :		<None>									
G01	1	WELDNECK	100		900 23.13	5.45	WN	1.00			No
Tag No. :		<None>									
G01	2	WELDNECK	100		900 23.13	5.45	WN	1.00			No
Tag No. :		<None>									
I01	1	WELDNECK	100		900 23.13	5.45	WN	1.00			No
Tag No. :		<None>									
I01	2	WELDNECK	100		900 23.13	5.45	WN	1.00			No

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REDUCER DATA LISTING									
From	To	Length	OD	OD	Thk	Thk	From	Cone	SIF
Point	Point	(mm)	From	To	From	To	Material	Angle	
			(mm)	(mm)	(mm)	(mm)		(deg)	
B03	B04	620.00	914.4	1016.0	23.8	24.0	5LX-X70		2.00t

t = Use SIF of 2.0

C E N T E R O F G R A V I T Y R E P O R T

Description		Weight (kg)	X CoG (mm)	Y CoG (mm)	Z CoG (mm)
-----		-----	-----	-----	-----
Valve	:	24190.49	23564.84	7500.00	-760.00
Support	:	0.00	0.00	0.00	0.00
Flange	:	29022.70	25859.53	7511.16	-692.89
Flexible Joint	:	0.00	0.00	0.00	0.00
Additional Weight	:	1678.30	9000.00	7500.00	0.00
Pipes (Run, Bend, Tee, Reducer)	:	328213.09	-248668.0	5249.88	-89.65
PipeTot (Valve+Support+Flange+Flex. Jt.+Add. Wt.+Pipes):		383104.47	-209552.4	5573.13	-177.28
Insulation	:	0.00	0.00	0.00	0.00
Cladding	:	0.00	0.00	0.00	0.00
Lining	:	0.00	0.00	0.00	0.00
Contents	:	0.00	0.00	0.00	0.00
Beam	:	0.00	0.00	0.00	0.00
PipeTot + Ins.	:	383104.47	-209552.4	5573.13	-177.28
PipeTot + Ins. + Clad.	:	383104.47	-209552.4	5573.13	-177.28
PipeTot + Ins. + Clad. + Lin.	:	383104.47	-209552.4	5573.13	-177.28
PipeTot + Ins. + Clad. + Lin. + Cont.	:	383104.47	-209552.4	5573.13	-177.28
PipeTot + Beam	:	383104.47	-209552.4	5573.13	-177.28
PipeTot + Ins. + Beam	:	383104.47	-209552.4	5573.13	-177.28
PipeTot + Ins. + Clad. + Beam	:	383104.47	-209552.4	5573.13	-177.28
PipeTot + Ins. + Clad. + Lin. + Beam	:	383104.47	-209552.4	5573.13	-177.28
PipeTot + Ins. + Clad. + Lin. + Cont. + Beam	:	383104.47	-209552.4	5573.13	-177.28

NOTE: Fluid Density Factor is assumed as 1.0.

NOTE: COG report only includes the currently visible beams and segments.

All pipe segments are included.

No beams present in the model.

L O A D C A S E		D E S C R I P T I O N
Load Case	Load Case Name	Description
GR	Gravity	
T1	Thermal 1	
T2	Thermal 2	
T3	Thermal 3	
W1	Wind 1	
W2	Wind 2	
W3	Wind 3	
W4	Wind 4	
P1	Press 1	
P2	Press 2	
P3	Press 3	

HYDROTEST LOAD SUMMARY

Specific Gravity of Fluid : 1.000

Hydrotest Pressure Case : P1

Hydrotest Pressure Factor : 1.250

Hydrotest Temperature Case : T3

Hydrotest Temperature Factor : 1.000

Cut-short Case : None

Include Insulation & Cladding: No

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ADDITIONAL WEIGHT DATA LISTING

Point Name	Added Weight kg	DX mm	DY mm	DZ mm
A07	1678.30			

Point Summary

Number of Points in the System				Mass	Weight of System(kg)		
Pipe	Frame	Soil	Total		Pipe	Contents	Total
77	0	1126	1203	0	383104.5	0.0	383104.5

ANCHOR DATA LISTING

Point Name	Translational Stiffness			Rotational Stiffness			Release Hanger		
	KTX N/mm	KTY N/mm	KTZ N/mm	KRX N.m/deg	KRY N.m/deg	KRZ N.m/deg	X XX	Y YY	Z ZZ
A00	RIGID	RIGID	RIGID	Connected to: Ground		RIGID			

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	DX	Offset (mm) DY	DZ	Global (mm) X	Y	Z
*** Segment A										
Origin	A00	Point	36"CL900	0.00	-562000.00	4940.00	-0.02	-562000.00	4940.00	-0.02
A00	A01	Run	36"CL900	46500	558000.00	0.00	0.02	-4000.00	4940.00	0.00
A01	A02	Bend	36"CL900	166.67	2000.00	0.00	0.00	-2000.00	4940.00	0.00
A02	A03	Run	36"CL900	174.99	2000.00	640.00	0.00	0.00	5580.00	0.00
A03	A04	Run	36"CL900	174.99	2000.00	640.00	0.00	2000.00	6220.00	0.00
A04	A05	Bend	36"CL900	349.98	4000.00	1280.00	0.00	6000.00	7500.00	0.00
A05	A06	Run	36"CL900	125.00	1500.00	0.00	0.00	7500.00	7500.00	0.00
A06	A07	Run	36"CL900	125.00	1500.00	0.00	0.00	9000.00	7500.00	0.00
A07	A08	Run	36"CL900	216.67	2600.00	0.00	0.00	11600.00	7500.00	0.00
A08	A09	Valve	36"CL900	190.50	2286.00	0.00	0.00	13886.00	7500.00	0.00
A09	A10	Tee	36"CL900	274.73	3296.80	0.00	0.00	17182.80	7500.00	0.00
A10	A11	Run	36"CL900	100.00	1200.00	0.00	0.00	18382.80	7500.00	0.00
A11	A12	Valve	36"CL900	190.50	2286.00	0.00	0.00	20668.80	7500.00	0.00
A12	A13	Run	36"CL900	274.73	3296.80	0.00	0.00	23965.60	7500.00	0.00
A13	A14	Valve	36"CL900	190.50	2286.00	0.00	0.00	26251.60	7500.00	0.00
A14	A15	Tee	36"CL900	83.07	996.80	0.00	0.00	27248.40	7500.00	0.00
A15	A16	Run	36"CL900	216.67	2600.00	0.00	0.00	29848.40	7500.00	0.00
A16	A17	Run	36"CL900	104.17	1250.00	0.00	0.00	31098.40	7500.00	0.00
A17	A18	Run	36"CL900	104.17	1250.00	0.00	0.00	32348.40	7500.00	0.00
A18	A19	Run	36"CL900	204.17	2450.00	0.00	0.00	34798.40	7500.00	0.00
A19	A20	Valve	36"CL900	190.50	2286.00	0.00	0.00	37084.40	7500.00	0.00
A20	A21	Run	36"CL900	12.77	-153.20	0.00	0.00	36931.20	7500.00	0.00
*** Segment B										
Origin	A21	Point	36"CL900	0.00	36931.20	7500.00	0.00	36931.20	7500.00	0.00
A21	B01	Junction	36"CL900	108.33	1300.00	0.00	0.00	38231.20	7500.00	0.00

From	To	Comp.	Pipe Id	Length mm	Offset (mm)			Global (mm)		
					DX	DY	DZ	X	Y	Z
B01	B02	Tee	36"CL9001	116.67	1400.00	0.00	0.00	39631.20	7500.00	0.00
B02	B03	Run	36"CL9003	333.33	4000.00	0.00	0.00	43631.20	7500.00	0.00
B03	B04	Reducer	40"PIPE	51.67	620.00	0.00	0.00	44251.20	7500.00	0.00
B04	B05	Tee	40"PIPE	50.00	600.00	0.00	0.00	44851.20	7500.00	0.00
B05	B06	Tee	40"PIPE	241.67	2900.00	0.00	0.00	47751.20	7500.00	0.00
B06	B07	Run	40"PIPE	141.67	1700.00	0.00	0.00	49451.20	7500.00	0.00
B07	B08	Run	40"PIPE	100.00	1200.00	0.00	0.00	50651.20	7500.00	0.00
*** Segment C										
Origin	B05	Tee	12"C06	0.00	44851.20	7500.00	0.00	44851.20	7500.00	0.00
B05	C01	Run	12"C06	58.33	0.00	700.00	0.00	44851.20	8200.00	0.00
*** Segment F										
Origin	A15	Tee	36"CL900	0.00	27248.40	7500.00	0.00	27248.40	7500.00	0.00
A15	F01	Bend	36"CL9002	233.33	0.00	2800.00	0.00	27248.40	10300.00	0.00
F01	F02	Bend	36"CL9003	16.67	0.00	0.00	-3800.00	27248.40	10300.00	-3800.00
F02	F03	Bend	36"CL9002	233.33	0.00	-2800.00	0.00	27248.40	7500.00	-3800.00
F03	F04	Run	36"CL9001	133.33	-1600.00	0.00	0.00	25648.40	7500.00	-3800.00
F04	F05	Valve	36"CL9001	190.50	-2286.00	0.00	0.00	23362.40	7500.00	-3800.00
F05	F06	Run	36"CL9003	333.33	-4000.00	0.00	0.00	19362.40	7500.00	-3800.00
F06	F07	Bend	36"CL9001	181.63	-2179.60	0.00	0.00	17182.80	7500.00	-3800.00
F07	F08	Bend	36"CL9002	233.33	0.00	2800.00	0.00	17182.80	10300.00	-3800.00
F08	F09	Bend	36"CL9003	16.67	0.00	0.00	3800.00	17182.80	10300.00	0.00
F09	A10	Tee	36"CL9002	233.33	0.00	-2800.00	0.00	17182.80	7500.00	0.00
*** Segment G										
Origin	B02	Tee	4"C06	0.00	39631.20	7500.00	0.00	39631.20	7500.00	0.00
B02	G01	Run	4"C06	58.33	0.00	700.00	0.00	39631.20	8200.00	0.00
*** Segment I										

Point Connectivity

From	To	Comp.	Pipe Id	Length mm	Offset (mm)			Global (mm)		
					DX	DY	DZ	X	Y	Z
Origin	B06	Tee	4"C06	0.00	47751.20	7500.00	0.00	47751.20	7500.00	0.00
B06	I01	Run	4"C06	58.33	0.00	700.00	0.00	47751.20	8200.00	0.00

